



Editorial

PLS-SEM: Looking Back and Moving Forward



This article introduces and motivates an exchange of thoughts on the paper by Edward E. Rigdon in the first of two *Long Range Planning* special issues on partial least squares structural equation modeling (PLS-SEM) in strategic management published in 2012 and 2013. For 30 years, there has been a heated debate on the benefits and drawbacks of PLS-SEM versus those of its sibling, the covariance-based structural equation modeling (CB-SEM) approach. Edward E. Rigdon's paper is a milestone that proposes a change of thought and encourages the long-required emancipation of the PLS-SEM method from CB-SEM. These developments will have a pronounced impact on the proper application of SEM as a key multivariate analysis method in the strategic management discipline, further enhancing the potential it has as a research tool.

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Introduction

Under the general theme “looking back and moving forward,” *Long Range Planning* initiates an exchange of comments, which starts with three invited papers by Peter M. Bentler and Wenjing Huang (2014), *On Components, Latent Variables, PLS and Simple Methods: Reactions to Rigdon's Rethinking of PLS*, Theo K. Dijkstra (2014), *PLS' Janus Face*, and Sarstedt et al. (2014a), *On the Emancipation of PLS-SEM*. These articles follow the common theme of this scientific discourse in that they reflect back and constructively launch partial least squares structural equation modeling (PLS-SEM) as an established, independent method. The first two articles deal with the methodological foundations of PLS-SEM as a component-based method and, amongst others, address ways to handle the well known consistency issues inherent in the method. Even critiques of PLS-SEM (e.g., McIntosh et al., 2014) call the consistent PLS algorithms such as the PLSe2 (Bentler et al., 2014) and PLSc (Dijkstra, 2014) “impressive”.

While these two articles (Bentler et al., 2014; Dijkstra, 2014) are more technical in nature, the third article (Sarstedt et al., 2014a) sheds further light on specific subject areas. Rigdon (2012) discusses by commenting on the distinction between explanatory modeling and predictive modeling, and the idea of model fit and its implications for SEM users from an application-oriented perspective. Finally, Rigdon (2014) responds to the comments in his article (“*Rethinking Partial Least Squares Path Modeling: Breaking Chains and Forging Ahead*”).

Looking back

The extent to which an issue is raised by successive generations of researchers and practitioners is an indicator of its importance. The benefits and limitations of PLS-SEM are issues that have long been heatedly debated across a wide variety of disciplines. The debate started with the development of PLS-SEM by Herman Wold (1974, 1982) and covariance-based SEM (CB-SEM) by Karl G. Jöreskog (1978, 1982). Composite-based PLS-SEM and factor-based CB-SEM were developed as complementary, but different, statistical methods with distinctive goals and requirements. In the early 1980s, the founders of both methods (Jöreskog et al., 1982) clearly emphasized these issues in their groundbreaking article “*The ML and PLS Techniques For Modeling with Latent Variables: Historical and Comparative Aspects*” with additional comments later (e.g., Dijkstra, 1983). However, CB-SEM was initially established as the primary method for estimating and testing structural equation modeling in the social science disciplines (e.g., Babin et al., 2008; Shah and Goldstein, 2006; Shook et al., 2004). Although both approaches were developed at about the same time, the initial dominance of CB-SEM is likely linked to the LISREL software, which was already available in the 1970s (e.g., Jöreskog and Sörbom, 1972), and methodological advances and applications have flourished since. In contrast, the PLS-SEM alternative was seldom recognized or used initially, much less improved or extended. The most notable exception was Jan-Bernd Lohmöller, who worked on the method continuously and wrote the first

comprehensive statistical textbook on PLS-SEM (Lohmöller, 1989), as well as the LVPLS program (Lohmöller, 1984). The textbook was not widely recognized, however, and the FORTRAN program was difficult to obtain and use.

Wynne W. Chin's (1998) scholarly work and the availability of graphical user interfaces for the LVPLS program (e.g., PLS Graph; Chin, 2003) marked the comeback of the PLS-SEM method. With clear-cut guidelines on how to use the method, how to evaluate and interpret the results, and how to run the analysis in a software application with an easy to apply user interface, the PLS-SEM method became broadly accessible. As a consequence, PLS-SEM has been increasingly adopted by social science disciplines (e.g., Hair et al., 2012b; Hair et al., 2012c; Lee et al., 2011; Peng and Lai, 2012; Ringle et al., 2012), experiencing increasing dissemination in various sub-disciplines (e.g., family business research; Binz Astracan et al., 2014; Sarstedt et al., 2014b). After special issues on the PLS-SEM method appeared in journals such as *MIS Quarterly* (Marcoulides et al., 2009) and the *Journal of Marketing Theory and Practice* (Hair et al., 2011b), *Long Range Planning* – as a primary journal in the strategic management field – devoted two special issues (Hair et al., 2012a, 2013; Robins, 2012) and this follow-up to it. Furthermore, PLS-SEM serves as a basis for estimating two of the most prominent and important models in marketing and MIS research; the American customer satisfaction index (ACSI) model (Fornell et al., 1996) and the technology acceptance model (TAM; Davis, 1989). Moreover, new software applications (e.g., SmartPLS; Ringle et al., 2005) and first textbooks for applied research (e.g., Hair et al., 2014) have been published.

Today, there is little doubt that the development of PLS-SEM – independent from any legitimate criticism of the method – is a success story. Various methodological advances have contributed to PLS-SEM's success. These advances include confirmatory tetrad analysis (CTA-PLS) to empirically assess the measurement model type (i.e., formative or reflective; Gudergan et al., 2008), importance-performance matrix analysis (IPMA) of PLS-SEM results (e.g., Höck et al., 2010; Rigdon et al., 2011; Völckner et al., 2010), approaches to assess hierarchical component models (e.g., Becker et al., 2012; Kuppelwieser and Sarstedt, 2014; Ringle et al., 2012; Wetzels et al., 2009), PLS-SEM-specific data segmentation techniques (e.g., Becker et al., 2013; Kuppelwieser and Sarstedt, 2014; Rigdon et al., 2010; Rigdon et al., 2011; Sarstedt, 2008; Sarstedt et al., 2011a; Sarstedt and Ringle, 2010), analysis of interaction effects (Henseler and Chin, 2010; Henseler and Fassott, 2010), nonlinear effects (Dijkstra and Henseler, 2011; Henseler et al., 2012; Rigdon et al., 2010) or multi-group analysis procedures (e.g., Rigdon et al., 2010; Sarstedt et al., 2011b).

Moving forward

With the increasing success of PLS-SEM, the critics lined up. One line of arguments examined the supposed misapplications of PLS-SEM as they relate to the typical arguments in favor of PLS-SEM (small sample sizes, less restrictive distributional assumptions, large model complexity, less restrictive use of formative measurement models). For example, the *MIS Quarterly* editorial by George A. Marcoulides and Carol Saunders (2006) appears to criticize the PLS-SEM method by focusing on the small sample size argument. Proponents of the PLS-SEM method have been anxious to engage in these discussions (e.g., Hair et al., 2011a; Henseler et al., 2009) but view them from a different perspective. For example, researchers' mistakes in collecting appropriate samples are not a deficiency unique to the PLS-SEM method. Moreover, there is no particular magic in PLS-SEM – as in any other multivariate analysis method – to overcome such shortcomings. In fact, most PLS-SEM users would try to obtain relatively large samples when the goal is to mimic CB-SEM (i.e., consistency at large; Wold, 1982). When the sample size is small, however, PLS-SEM has an inherent advantage in that it is robust and exhibits relatively higher statistical power (Reinartz et al., 2009). In contrast, CB-SEM in similar situations has several notable weaknesses, including a lack of robustness (Boomsma and Hoogland, 2001). The obvious misapplications of PLS-SEM in many past research articles resulted in the publications of guidelines for researchers, reviewers and editors on how to conduct PLS-SEM studies (e.g., Chin, 2010; Hair et al., 2011a, 2013; Hair et al., 2012c).

Other researchers have focused on the inability of PLS-SEM to mimic CB-SEM. For example, McDonald (1996) proposed several criticisms to which Dijkstra (2010) persuasively replies. A final stream of criticism uses simulation studies to examine alleged deficiencies in the performance of PLS-SEM. For example, Hwang et al. (2010) claim the inferiority of PLS-SEM based on the results of a limited simulation study published in the *Journal of Marketing Research*. But Henseler (2012), in his response published in the *Journal of the Academy Science*, points out several serious flaws that led to the incorrect conclusions of Hwang et al. (2010). At the same time, a comprehensive simulation study by Reinartz et al. (2009) in the *International Journal of Research in Marketing* shows that PLS-SEM performs fairly well in mimicking CB-SEM and clearly substantiates the well-known advantages and disadvantages of each method.

The discussion of PLS-SEM's ability to mimic CB-SEM surfaced again more recently (Goodhue et al., 2012a, 2012b; Marcoulides et al., 2012) and is ongoing. Academic neutrality in these discussions is not always a given: sometimes a term with a negative connotation is declared to be a characteristic of PLS-SEM, stretching the meaning of the term beyond recognition, and ignoring the fact that in its non-standard meaning it applies to all methods, whatever their pedigree. A case in point is 'capitalization on chance' that in statistical methodology refers to the phenomenon that models tend to be changed or selected in the light of the data, and to the need and notorious difficulty to honestly and properly account for that in statistical inference. As applied to PLS-SEM however, 'capitalization on chance' typically refers to certain small sample tendencies, also known as small sample bias, which has a much less alarming ring to it. Every non-linear estimator must by necessity display some bias, this cannot be a surprise. What is a surprise is that the negative term tends to be used especially when the 'bias' is actually helpful, and the expected value of the PLS-SEM estimator in small samples is closer to the true value than its probability limit (its value in very large samples) would predict. One vivid example of how non-constructive and misguided these

Criterion	PLS-SEM	CB-SEM
Objective:	Prediction oriented	Parameter oriented
Approach:	Variance based	Covariance based
Assumptions:	Predictor specification (nonparametric)	Typically multivariate normal distribution and independent observations (parametric)
Parameter estimates:	Consistent as indicators and sample size increase (i.e., consistency at large)	Consistent
Latent variable scores:	Explicitly estimated	Indeterminate
Epistemic relationship between a latent variable and its measures:	Can be modeled in either formative or reflective mode	Typically only with reflective indicators
Implications:	Optimal for prediction accuracy	Optimal for parameter accuracy
Model complexity:	Large complexity (e.g., 100 constructs and 1000 indicators)	Small to moderate complexity (e.g., less than 100 indicators)
Sample size:	Power analysis based on the portion of the model with the largest number of predictors. Recommendations for the minimum number of observations range from 30 to 100 cases.	Ideally based on power analysis of specific model. Recommendations for the minimum number of observations generally range from 200 to 800.

Figure 1. Comparison of PLS-SEM and CB-SEM (Chin and Newsted, 1999)

discussions can become was presented by Rönkkö et al. (2013; p. 443), who set out to dispel “statistical myths and urban legends surrounding the often-stated capabilities of the PLS method.” The article by Rönkkö et al. (2013) was nothing short of a polemic, including misrepresentations of the literature. In a recent rejoinder, Henseler et al. (2014) demonstrate that the alleged weaknesses of PLS are not inherent in the method at all but are rather the result of the limitations in Rönkkö et al. (2013) study and its questionable research designs for the analysis of the performance of the PLS-SEM method.

Moving forward does not mean overlooking qualified criticism. However, one has to keep in mind that PLS-SEM was not designed to perfectly mimic CB-SEM. If scholars are expecting the method to mimic CB-SEM, they must work within the constraints of the proposed disadvantages, which are often not as serious as claimed, while exploiting some advantages. In moving forward, it is important to establish composite-based PLS-SEM as a method of its own, with uniquely specified advantages. Chin and Newsted (1999) continued the initial line of thought established by Jöreskog and Wold (1982) when reviving PLS-SEM. They tried to distinguish PLS-SEM from CB-SEM as an independent method (Figure 1). However, instead of recognizing the difficulties when comparing the two different statistical methods, researchers interpreted PLS-SEM and CB-SEM as two competing methods, which aim at achieving the same objective. More specifically, in the tradition of Fornell and Bookstein (1982), they simply regarded PLS-SEM as an alternative that mimics factor-based CB-SEM.¹ Rigdon (2012) advocates the new goal to emancipate composite-based SEM (e.g., PLS-SEM) as a method for estimating complex cause-effect relationship models.

A first step towards the establishment of PLS-SEM as a composite-based SEM method has been presented by Rigdon (2013) and Henseler et al. (2014). Following Rigdon (2012), additional steps of rethinking the PLS-SEM method and moving forward by constructively advancing and emancipating the PLS-SEM method imply several beneficial areas. From our point of view, the following areas are of key relevance:

- For what kind of studies is the composite-based (i.e., PLS-SEM) approach better suited than the factor-based (i.e., CB-SEM) approach?
- What kind of procedure should researchers follow to establish a composite-based (i.e., PLS-SEM) model?
- What precisely does exploratory research and prediction-orientation mean in the context of composite-based (i.e., PLS-SEM) studies?
- Can bias correction of composite-based (i.e., PLS-SEM) model estimations perfectly mimic CB-SEM results?
- Should global goodness-of-fit criteria be developed for the composite-based (i.e., PLS-SEM) approach?

¹ In this kind of comparison, it was common to stress some of PLS-SEM's advantageous features by its proponents (e.g., greater flexibility regarding the distribution of data, handling of complex models, relatively unrestricted use of formative measurement models, and robust results with small sample sizes). Critics argue against these advantages, reveal additional disadvantageous features and sometimes fully condemn the use of PLS-SEM.

- How would such global goodness-of-fit criteria affect the use of the composite-based (i.e., PLS-SEM) approach? Would it also follow the maximum fit paradigm of the factor-based (i.e., CB-SEM) approach or would it follow other goals or combinations of goals?

Taking this wider spectrum of structural equation modeling techniques into account, the most salient relative advantage of consistent PLS algorithms such as the PLSe2 (Bentler and Huang, 2014) and PLSc (Dijkstra, 2014) is their ability to build a bridge between factor models and composite models. So far, variance-based SEM methods are unable to consistently estimate factor models (Henseler, 2012; McDonald, 1996), and covariance-based SEM methods have difficulties modeling endogenous composites (Rigdon, 2013). Consistent PLS methods are readily equipped to fill this gap and models that simultaneously involve factors and composites are likely to become a fruitful field of their application. Whereas the correction of consistent PLS methods (e.g., PLSc) would be applied to those constructs that are modeled as common factors, uncorrected traditional PLS-SEM would be applied to those constructs that are modeled as composites.

In the course of emancipating PLS-SEM from CB-SEM, we envision many research opportunities. Knowledge from both research streams would be highly beneficial when following this kind of direction. In fact, PLS-SEM would likely benefit from the 30 years of methods development and improvements that have been conducted for CB-SEM. Using this knowledge enables researchers to avoid dead ends and focus on the most crucial topics. Hence, we envision an increasingly dynamic development of the PLS-SEM method in the near future.

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Biographies

Dr. Marko Sarstedt is Professor of Marketing at Otto-von-Guericke-University Magdeburg, Germany, and Visiting Professor at the University of Newcastle, Australia. His research interests include PLS-SEM, measurement principles, and corporate reputation. His research has been published in journals such as the *Journal of the Academy of Marketing Science*, *International Journal of Research in Marketing*, *Organizational Research Methods*, *Journal of World Business*, and *MIS Quarterly*.

Please visit <http://www.marketing.ovgu.de/marketing/en/Marketing.html> for more information on Dr. Sarstedt. E-mail: marko.sarstedt@ovgu.de

Dr. Christian M. Ringle is a Professor of Management and Director of the Institute of Human Resource Management and Organizations (HRMO) at Hamburg University of Technology (TUHH), Germany, and he is Visiting Professor to the Faculty of Business and Law Professor at the University of Newcastle, Australia. His research mainly addresses strategic management, organizations, marketing, human resource management, and quantitative methods for business and market research. Please visit <http://www.tuhh.de/hrmo> for more information on Dr. Ringle. E-mail: c.ringle@tuhh.de

Dr. Joseph F. Hair is Professor of Marketing at Coles College of Business, Kennesaw State University, USA. His research mainly focuses on multivariate analysis methods and their application in business research. Please visit http://coles.kennesaw.edu/departments_faculty/faculty-pages/Hair-JoeF.htm for more information on Dr. Hair. E-mail: jhair3@kennesaw.edu

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